

KILIMANJARO INTL, Tanzania, United Republic of

WMO: 637910

Lat: 3.429S

Lon: 37.074E

Elev: 2932

StdP: 13.20

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	56.1	58.6	47.4	53.9	83.4	49.7	58.8	80.0	18.6	77.6	16.8	76.9	0.8	330	0.329

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.9	94.6	67.9	93.0	68.1	91.5	68.5	72.9	83.5	72.1	82.9	71.6	82.6	11.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.2	124.1	75.2	69.8	122.5	74.5	69.1	119.3	74.0	38.7	83.6	38.0	83.0	37.4	82.1	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	20.7	18.3	DB	51.4	97.8	1.9	2.1	50.1	99.3	49.0	100.6	47.9	101.7	46.6	103.3
			WB	49.3	77.0	3.0	2.0	47.1	78.4	45.4	79.6	43.7	80.7	41.5	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.4	79.0	79.5	79.6	77.0	73.3	70.5	69.8	71.2	73.2	76.3	78.0	78.2	(6)
	DBStd	4.20	2.09	2.08	2.47	2.58	2.26	2.49	2.34	2.19	2.43	2.53	1.99	1.98	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	(9)
	CDD50	9284	898	827	918	810	722	615	614	656	695	814	840	875	(10)
	CDD65	3809	433	407	453	360	257	165	149	191	245	349	390	410	(11)
	CDH74	30447	4416	4154	4197	2348	1063	712	734	1112	1920	2959	3237	3595	(12)
	CDH80	11888	2102	1999	1944	780	162	66	66	169	569	1184	1338	1509	(13)
(14) Wind	WSAvg	7.0	9.7	9.9	8.2	5.5	3.9	3.9	4.3	5.6	7.1	8.3	8.8	8.8	(14)
(15) Precipitation	PrecAvg	36.7	1.8	1.6	4.1	10.9	7.3	1.5	1.0	0.8	0.6	1.4	2.8	2.7	(15)
	PrecMax	54.0	7.2	6.0	11.2	19.6	11.8	4.7	1.9	3.1	4.5	6.3	8.5	7.4	(16)
	PrecMin	24.8	0.0	0.0	0.9	3.5	1.6	0.6	0.3	0.2	0.1	0.1	0.4	0.0	(17)
	PrecStd	8.5	1.8	1.4	2.2	4.4	2.4	0.7	0.3	0.6	0.9	1.5	1.8	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.4	96.3	96.0	91.8	86.3	84.1	84.2	86.2	89.4	92.1	92.5	93.4	(19)
		MCWB	66.7	68.5	69.3	69.8	69.3	67.4	65.2	65.4	66.0	66.6	68.0	67.2	(20)
	2%	DB	93.6	94.6	93.6	89.4	84.1	81.9	81.6	84.0	87.5	90.0	91.1	91.6	(21)
		MCWB	67.5	68.0	68.7	69.7	68.9	66.5	64.6	64.7	65.9	66.5	68.0	68.1	(22)
	5%	DB	91.8	92.5	91.7	87.1	82.0	80.4	80.3	82.2	85.8	88.2	89.4	89.9	(23)
		MCWB	68.2	68.4	69.1	69.6	68.7	65.9	64.3	64.2	65.1	66.6	68.1	68.7	(24)
	10%	DB	90.1	90.6	89.8	84.5	80.3	78.5	78.5	80.5	84.0	86.7	87.7	88.0	(25)
		MCWB	68.5	68.6	69.0	70.1	68.5	65.7	63.9	64.2	64.8	66.5	68.4	69.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	72.7	73.5	73.8	72.0	69.3	68.2	68.2	69.3	71.8	73.3	73.8	(27)
		MCDB	85.2	85.6	85.8	82.2	78.5	78.2	79.4	81.0	82.6	82.4	84.0	83.7	(28)
	2%	WB	71.7	71.8	72.3	72.6	71.0	68.3	66.5	66.7	67.9	70.1	71.7	72.2	(29)
		MCDB	84.9	85.2	84.6	81.4	77.9	77.1	77.0	78.6	80.9	80.3	82.8	82.7	(30)
	5%	WB	70.9	70.9	71.6	71.8	70.2	67.4	65.6	65.8	66.9	69.1	70.7	71.2	(31)
		MCDB	84.0	84.2	83.8	80.4	77.2	75.9	75.7	77.2	79.4	79.6	81.7	82.1	(32)
	10%	WB	70.1	70.2	70.9	71.1	69.4	66.6	64.7	65.0	66.0	68.3	69.9	70.5	(33)
		MCDB	83.2	83.2	82.6	79.5	76.3	74.5	74.7	76.4	78.3	79.1	80.8	81.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	24.1	23.9	21.7	16.5	14.2	16.9	18.7	19.6	22.4	21.9	20.9	21.4	(35)
		MCDBR	26.9	26.8	24.5	19.6	16.9	19.7	21.5	22.6	25.8	24.4	23.1	23.9	(36)
	5% WB	MCWBR	7.1	6.9	6.0	5.1	5.3	7.4	8.3	7.6	8.4	6.7	5.9	6.5	(37)
		MCWBR	22.4	23.1	20.6	16.4	13.9	16.9	18.8	19.6	22.2	20.8	20.0	20.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.452	0.489	0.451	0.376	0.369	0.414	0.439	0.428	0.409	0.406	0.403	0.434	(40)
		taud	2.241	2.115	2.262	2.513	2.511	2.337	2.223	2.259	2.317	2.340	2.390	2.319	(41)
	Ebn at Noon		281	271	278	292	286	268	263	273	286	291	293	285	(42)
		Edh at Noon	47	53	45	34	33	38	43	43	43	42	40	43	(43)
(44) All-Sky Solar Radiation	RadAvg		2003	2060	1945	1676	1455	1386	1425	1567	1852	1938	1840	1884	(44)
	RadStd		130	122	138	99	66	67	125	112	109	106	93	106	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page